

Full Stack Development Summary

Covered Topics:

- INTRODUCTION OF JAVA
- OVERVIEW OF PROGRAMMING LANGUAGE
- TYPES OF PROGRAMMING LANGUAGE
- HISTORY OF JAVA
- WHY WE LEARN JAVA
- COMPILER AND INTERPRETER
- FEATURE OF JAVA
- JVM (JAVA VIRTUAL MACHINE)
- JRE (JAVA RUNTIME ENVIRONMENT)
- JDK (JAVA DEVELOPMENT KIT)
- IDE (INTEGRATED DEVELOPMENT ENVIRONMENT)
- MAIN FEATURES
- JAVA'S MAGIC BYTECODE

INTRODUCTION OF JAVA

Java is a truly Object Oriented Programming Language . Java was created by James Gosling and his team SunMicrosystem in 1991. Java's old name was Oak and later renamed as Java in 1995(Oracle). Java is a reusable , Secure and Platform Independent . Java have core principle "Write Once, Run Anywhere".

OVERVIEW OF PROGRAMMING LANGUAGE

Programming language is a communication between human and Machine.

TYPES OF PROGRAMMING LANGUAGE

Low Level Language - Assembly Code (Mnemonic code) , Chips.

Medium Level Language - C,C++.

High Level Language – Java , Python.

HISTORY OF JAVA

- 1991- Java created by James Gosling . language is called Oak.
- 1995 – later rename as Java.
- 2014- every six month cycle to update the version.
- Latest version of Java is 26.
- Java include JDK,JRE and JVM.

WHY WE LEARN JAVA

Java is a robust , Secure and Platform Independent. And we use java in many apps like android, Web Application , Finance and Banking, etc,.....

COMPILER AND INTERPRETER

COMPILER: Compiler is to find a syntax error and source code (.java file) to convert bytecode(.class file) .

INTERPRETER: JVM is use the principle” WRITE ONCE RUN ANYWHERE”. Interpreter use JIT for speed and its convert byte code into machine code. Its auto check error in line by line .

JVM(JAVA VIRTUAL MACHINE)

- JVM is a important to run Java Programs.
 - JVM convert bytecode into machine code
- JVM Does(work):
- Loads Code
 - Verifies Code
 - Execute Code – converts Bytecode into Machine Code using:

Interpreter

JIT(Just In Time compiler)

- Manage Memory

JRE(JAVA RUNTIME ENVIRONMENT)

- It Include JVM & Libraries(pre – built code).
- JVM is using Javac compiler to compile the program
- Inside JRE:

JVM

Core Libraries(java.util, java.io,java.lang)

Supporting files

- Example: Run Java program but not write or compile programs

JDK(JAVA DEVELOPMENT KIT)

- JDK include Write, compile, debug , and run java programs
- Its Inside :

JRE

JVM+ Core Libraries

- Development Tools:

Compiler (javac)

Debugger (jdb)

Documentation tool (Javadoc)

INTEGRATED DEVELOPMENT ENVIRONMENT

- Popular IDEs for Java:

Eclipse

IntelliJ IDEA

NetBeans

VS Code

- Programmer's Workspace.
- Complete environment to write , edit, compile, debug, and run code easily.

MAIN FEATURES

1. Code Editor
2. Compile/Interpreter Integration
3. Debugger
4. Project Management

5. Other Tools (build tools, version control(Git), Plugin Support, Testing frameworks)

JAVA'S MAGIC BYTECODE

- If compile a program , becomes a .class file (bytecode)
- Every .class file starts with special code 0xCAFEBAFE.
- Secret Signature start of every java .class file. This is called Magic number.
- Its like a ID card of bytecode.
- When JVM opens a .class file first thing it checks whether it starts with special code.
- If its there – JVM Knows it's a valid Java bytecode files.
- If not – JVM rejects it (Prevents error).